Sandeep Banik

Postdoctoral Research Associate, University of Illinois Urbana-Champaign

sandeepbanik.github.io | baniksan@illinois.edu | +1-517-703-4044
Google Scholar | sandeepbanik | sandeepbanik

Research Interests

I develop game-theoretic and control-theoretic foundations for **principled authority allocation** among humans, robots, and AI agents: deciding when an autonomous system should act, defer, or hand over control, and with what guarantees. My work combines dynamic game theory, robust and adaptive control, and learning to build certifiable shared autonomy, adversarially resilient cyber-physical systems, and safe learning-enabled control.

Keywords: shared autonomy; human–robot and human–AI teaming; learning-enabled robot control; $\mathcal{L}_1$ adaptive and robust control; multi-robot and multi-agent coordination; dynamic and stochastic games; distributionally robust learning; safety and security of cyber-physical systems.

At a glance: 3 journal articles (IEEE TAC, IEEE T-RO, ACM TCPS) · 15 peer-reviewed conference papers (CDC, IROS, L4DC, RO-MAN, GameSec, ISIT, AIAA SciTech) · IEEE CSS TCSP Best Student Paper Award · contributor to 11 proposals, 2 funded (incl. $900K NSF CPS) · Technical Program Chair, GameSec 2026 · research mentor to 8 Ph.D. students

Education

2023 **Ph.D., Electrical and Computer Engineering**, Michigan State University, USA
Dissertation: *Adversarial Modeling in Game-Theoretic Frameworks for Securing Cyber-Physical Systems*
Advisor: Prof. Shaunak D. Bopardikar

2017 **M.Sc., Mechanical Engineering**, KU Leuven, Belgium
Thesis research with Siemens Industry Software NV and Ford Proving Grounds

2014 **B.Eng., Mechanical Engineering**, Visvesvaraya Technological University, India

Honors and Awards

2023 **Best Student Paper Award**, IEEE Control Systems Society Technical Committee on Security and Privacy, for [C9] (presented at CDC 2023)

2022 **Jackson Computing and Data Impact Award**, Michigan State University Research Symposium

Appointments and Research Experience

Postdoctoral Research Associate, University of Illinois Urbana-Champaign Jan 2024 – present
PI: Prof. Naira Hovakimyan

- *Shared autonomy:* cooperative takeover games that allocate authority between a human and an autonomous agent under stochastic human override; improved policy performance by 18.75% in simulation across diverse behavioral models [C1, C2].
- *Certifiable learning-enabled control:* co-developed L1-DS, a robust control layer for learned motion policies with provable tracking bounds on black-box robots (up to 88% lower tracking error than the nominal policy) [C3]; distributionally robust imitation learning with a priori robustness and performance guarantees [C4].
- *Safe aerial autonomy:* System-Theoretic Process Analysis (STPA) for vision-based VTOL landing and take-off, with MIT collaborators [C7]; potential-game multi-agent coordination [C6] and $\mathcal{L}_1$ adaptive moving-horizon estimation for UAVs [C5].
- *Research leadership:* contributed to 11 research proposals (2 funded); day-to-day research mentor to 8 Ph.D., 2 M.S., and 3 undergraduate researchers.

Graduate Research Assistant, Michigan State University Aug 2019 – Dec 2023
Dept. of ECE; Advisor: Prof. Shaunak D. Bopardikar

– Introduced `FlipDyn`, a class of dynamic games for resource takeovers in dynamical systems [J1, C9], and dynamic games with stopping states for attack-resilient motion planning [J3, C10, C11].
– Built adversary-in-the-loop evaluation and defense-planning pipelines with Pacific Northwest National Laboratory [J2, C12].
– Validated controllers in 50+ hardware experiments on ROS2 ground robots, including real-time leader–follower and lane-changing scenarios [C10]. [Demo 1], [Demo 2]

Ph.D. Intern, Machine Learning, Pacific Northwest National Laboratory Jun – Aug 2022
Advisor: Dr. Arnab Bhattacharya

– Integrated graph neural networks into multi-agent reinforcement learning for adversarial goal inference and coordinated mitigation (MARS research program) [J2].

Research Assistant, IIIT-Delhi, India Jul 2018 – Jul 2019
Advisors: Prof. Sanjit K. Kaul, Prof. P. B. Sujit

– Age-of-Information optimal sensor scheduling in gateway-based update systems [C14]; multi-vehicle path planning with Texas A&M University and LSTS, Portugal [C13, C15].

M.Sc. Thesis Researcher, KU Leuven, Belgium May 2016 – May 2017
With Siemens Industry Software NV and Ford Proving Grounds

– State and parameter estimation for vehicle testing: internal states of continuously variable transmissions, center of gravity and moment of inertia under varying loads, and extended Kalman filter–based fault detection for K&C rigs.

Publications

S. Banik in bold; † student mentored by S. Banik. Full list on Google Scholar.

Refereed Journal Articles

[J1] **S. Banik** and S. D. Bopardikar, "`FlipDyn` with Control: Resource Takeover Games with Dynamics," *IEEE Transactions on Automatic Control*, vol. 71, no. 3, pp. 1644–1659, Mar. 2026. doi: 10.1109/TAC.2025.3618811 [*link*]

[J2] **S. Banik**, T. Ramachandran, A. Bhattacharya, and S. D. Bopardikar, "Automated Adversary-in-the-Loop Cyber-Physical Defense Planning," *ACM Transactions on Cyber-Physical Systems*, vol. 7, no. 3, Article 18, 25 pp., Jul. 2023. doi: 10.1145/3596222 [*link*]

[J3] **S. Banik** and S. D. Bopardikar, "Attack-Resilient Path Planning Using Dynamic Games With Stopping States," *IEEE Transactions on Robotics*, vol. 38, no. 1, pp. 25–41, Feb. 2022. doi: 10.1109/TRO.2021.3123896 [*link*]

Refereed Conference Proceedings

[C1] **S. Banik** and N. Hovakimyan, "`Flip-Team`: Cooperative Takeover Games with Stochastic Human Override," *IEEE Conference on Decision and Control (CDC)*, 2026, to appear.

[C2] **S. Banik** and N. Hovakimyan, "Principled Authority Switching for Shared Autonomy in Human-Robot Teams," *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 2026.

[C3] E. Pathak†, A. Aboudonia, **S. Banik**, and N. Hovakimyan, "A Robust Task-Level Control Architecture for Learned Dynamical Systems," *Learning for Dynamics and Control Conference (L4DC)*, 2026.

[C4] A. Gahlawat, A. Aboudonia, **S. Banik**, N. Hovakimyan, N. Matni, A. D. Ames, G. Zardini, and A. Speranzon, "Distributionally Robust Imitation Learning: Layered Control Architecture for Certifiable Autonomy," *AIAA SCITECH Forum*, 2026.

[C5] T. Q. Nguyen†, M. Kim, **S. Banik**, J. Kim, and N. Hovakimyan, "Moving-Horizon State Estimation for Unmanned Aerial Vehicles: An $\mathcal{L}_1$ Adaptive Optimization Approach," *AIAA SCITECH Forum*, 2026.

[C6] M. Aramyan†, **S. Banik**, T. Bakaryan, and N. Hovakimyan, "PACT: Potential Differential Game-Based Multi-Agent Coordination for Time-Critical Missions," *AIAA SCITECH Forum*, 2026.

[C7] **S. Banik**, J. Kim, N. Hovakimyan, L. Carlone, J. Thomas, and N. G. Leveson, "Integrating Vision Systems and STPA for Robust Landing and Take-Off in VTOL Aircraft," *AIAA SCITECH Forum*, 2025, p. 1324. doi: 10.2514/6.2025-1324 [*link*]

[C8] **S. Banik**, S. D. Bopardikar, and N. Hovakimyan, "`FlipDyn` in Graphs: Resource Takeover Games in Graphs," *International Conference on Decision and Game Theory for Security (GameSec)*, Springer, 2024, pp. 220–239. doi: 10.1007/978-3-031-74835-6_11 [*link*]

[C9] **S. Banik** and S. D. Bopardikar, "`FlipDyn`: A Game of Resource Takeovers in Dynamical Systems," *IEEE Conference on Decision and Control (CDC)*, Cancun, Mexico, 2022, pp. 2506–2511. doi: 10.1109/CDC51059.2022.9992387 [*link*]
IEEE CSS TCSP Best Student Paper Award

[C10] **S. Banik** and S. D. Bopardikar, "Stochastic Games with Stopping States and their Application to Adversarial Motion Planning Problems," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Kyoto, Japan, 2022, pp. 13181–13188. doi: 10.1109/IROS47612.2022.9982011 [*link*]

[C11] **S. Banik** and S. D. Bopardikar, "Secure Route Planning Using Dynamic Games with Stopping States," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Las Vegas, NV, USA, 2020, pp. 2404–2409. doi: 10.1109/IROS45743.2020.9340884 [*link*]

[C12] A. Bhattacharya, T. Ramachandran, **S. Banik**, C. P. Dowling, and S. D. Bopardikar, "Automated Adversary Emulation for Cyber-Physical Systems via Reinforcement Learning," *IEEE International Conference on Intelligence and Security Informatics (ISI)*, Arlington, VA, USA, 2020, pp. 1–6. doi: 10.1109/ISI49825.2020.9280521 [*link*]

[C13] **S. Banik**, P. B. Sujit, and S. Rathinam, "Multi-AUV Coverage with Functional Constraints and Currents," *IEEE/OES Autonomous Underwater Vehicles Symposium (AUV)*, St. John's, NL, Canada, 2020, pp. 1–2. doi: 10.1109/AUV50043.2020.9267937 [*link*]

[C14] **S. Banik**, S. K. Kaul, and P. B. Sujit, "Minimizing Age in Gateway Based Update Systems," *IEEE International Symposium on Information Theory (ISIT)*, Paris, France, 2019, pp. 1032–1036. doi: 10.1109/ISIT.2019.8849367 [*link*]

[C15] **S. Banik**, S. Rathinam, and P. B. Sujit, "Min-Max Path Planning Algorithms for Heterogeneous, Autonomous Underwater Vehicles," *IEEE/OES Autonomous Underwater Vehicle Workshop (AUV)*, Porto, Portugal, 2018, pp. 1–6. doi: 10.1109/AUV.2018.8729714 [*link*]

Manuscripts Under Review

[S1] M. Jang†, T. Q. Nguyen†, V. Khatana, **S. Banik**, N. Hovakimyan, and P. Voulgaris, "Vision-Based Uncertainty-Aware Planning for Autonomous Aerial Refueling," submitted to *IEEE International Conference on Robotics and Automation (ICRA)*. (Title changed to maintain anonymity.)

[S2] **S. Banik** and S. D. Bopardikar, "Budget Based Stochastic Games with Stopping States," submitted to *IEEE Open Journal of Control Systems*.

[S3] **S. Banik**, S. Agarwal, S. Jha†, N. Hovakimyan, P. Zhao, and J. Wang, "Learning Takeover Policies for `Flip-Team`," submitted to *AAAI Conference on Artificial Intelligence, Special Track on AI Alignment*. (Title changed to maintain anonymity.)

Research Funding and Proposal Development

Proposals I contributed to writing as a postdoctoral researcher at UIUC.

Funded

2025 **CPS-FR: PIAS: Performativity-aware Intelligent Autonomous Systems**, NSF FutureCore: Cyber-Physical Systems (CPS-FR). **$900,000**.

2025 **REMIX: Robust Embodied Multimodal Interface eXchange**, HUJI. **$50,000**.

Submitted

2026 CPS-FR: Perception-Aware Planning with Distributionally Robust Safety Certificates, NSF FutureCore: CPS-FR

2025 Co-Adaptive Human–Robot Teaming in Teleoperated Aerial Manipulation, NSF M3X

2025 Intelligent Contingency Management for Advanced Air Mobility Platforms with Layered Architectures, NASA University Leadership Initiative (ULI)

2025 Architectures for Predictable and Certifiable Robust Autonomy, Collins Aerospace

2025 AI Safety for Inference-time Compute Models, Schmidt Sciences

2025 BRIDGE: Bi-directional Representation for Intent and Decision-making in Generative Environments, Sony Research Awards

2024 SAGE-AI: Synthesis and Adaptive Control of Generative AI, Amazon Research Awards

2024 A Control-theoretic Approach to Cognitive Modeling and Optimization of LLM-enhanced Human-agent Teams, Collective Judgment Formation Collaborative Research Alliance (CJF CRA) BAA

2024 STARLIGHT: System-Theoretic Approach for Risk-assured and Logic-driven IntelliGent autonomy and Human Teaming, Multidisciplinary University Research Initiative (MURI)

Talks and Presentations

2026 *`Flip-Team`: Cooperative Takeover Games with Stochastic Human Override*, IEEE RO-MAN

2026 *Shared Autonomy: From Robots to AI Agents*, Midwest Robotics Workshop (MWRW), Chicago, IL

2026 *Shared Autonomy: From Robots to AI Agents*, Coordinated Science Laboratory Student Conference, UIUC

2025 *Integrating Vision Systems and STPA for Robust Landing and Take-Off in VTOL Aircraft*, STAMP Workshop, MIT, March 2025

2024 *Transitioning from PhD: Roles and Opportunities*, Student Breakfast on Security and Privacy of Next-Generation Cyber-Physical Systems, American Control Conference (ACC)

2023 *Adversarial Modeling in Game-Theoretic Frameworks for Securing Cyber-Physical Systems*, Indraprastha Institute of Information Technology Delhi

2023 *A Game of Resource Takeovers for Dynamical Systems*, Midwest Workshop on Control and Game Theory

2023 *A Game of Resource Takeovers for Dynamical Systems*, CMSE Conference

2022 *Automated Adversary Emulation for Cyber-Physical Systems via Reinforcement Learning*, Michigan State University Research Symposium

Teaching

Teaching Experience

Fall 2026 **Teaching Assistant, ME 562: Robust Adaptive Control** (graduate), Department of Mechanical Science and Engineering, UIUC. Instructor: Prof. N. Hovakimyan. Derive proofs in lecture and lead MATLAB coding tutorials that pair each guarantee with a simulation that tests its assumptions.

Teaching Development

2025 **Scholarship and Teaching for Engineering Postdocs (STEP) program**, Grainger College of Engineering, UIUC: practice teaching with feedback, teaching observation, and workshops on evidence-based pedagogy; teaching-statement development.

Teaching Interests

Undergraduate	Control systems; signals and systems; dynamics; applied mathematics
Graduate	Linear systems theory; nonlinear control; optimization; robust and adaptive control; game theory
New electives	*Safe Learning-Enabled Control*; *Game-Theoretic Methods for Multi-Agent Systems*; *Interactive Autonomy*

Research Mentoring

Day-to-day research mentoring of students in the groups of Prof. N. Hovakimyan (UIUC) and Prof. S. D. Bopardikar (MSU). Outcomes refer to the publication list above.

Ph.D. Students (UIUC)

M. Aramyan	Potential-game multi-agent coordination → [C6] (first author)
E. Pathak	Robust control architectures for learned dynamical systems → [C3] (first author)
T. Q. Nguyen	Moving-horizon estimation for UAVs → [C5] (first author), [S1]; also as M.S. student
M. Jang	Vision-based uncertainty-aware planning → [S1] (first author, under review)
J. Bullock	Imitation learning with $\mathcal{L}_1$ distributionally robust adaptive control
S. Aich	Vision-based planning
M. Yeghiazaryan	STPA-based safety analysis
J. Li	Bayesian learning

M.S. Students

R. Madhavan	UIUC; contributed to the PACT framework [C6]
T. Q. Nguyen	UIUC; see Ph.D. students
A. Bodea	MSU; reinforcement learning projects
M. Viswanath	MSU; reinforcement learning projects

Undergraduate Researchers

UIUC	N. J. Wijas, S. Gummadapu, S. Jha (co-author, [S3])
MSU	A. Debaker, I. Kapale, B. Ganeshbabu

Professional Service and Outreach

Leadership and Organization

- **Technical Program Chair**, Conference on Game Theory and AI for Security (GameSec), 2026
- **Organizing Committee**, IEEE Control Systems Society Technical Committee on Security and Privacy (TCSP)
- **Session Co-Chair**, *Cognition, Intent & Embodied AI VIII*, IEEE RO-MAN 2026

Reviewing

Journals	IEEE Transactions on Automatic Control (TAC); IEEE Transactions on Robotics (T-RO); IEEE Transactions on Intelligent Transportation Systems (T-ITS); IEEE Transactions on Control of Network Systems (TCNS); IEEE Robotics and Automation Letters (RA-L)
Conferences	IEEE CDC; IEEE ACC; IEEE/RSJ IROS; IEEE ICRA; Robotics: Science and Systems (RSS); GameSec; IEEE ICCSN; Indian Control Conference

Outreach

- Presenter, Detroit Area Pre-College Engineering Program (DAPCEP), 2022–2023

References

Prof. Naira Hovakimyan (postdoctoral advisor)
W. Grafton and Lillian B. Wilkins Professor
Mechanical Science and Engineering
University of Illinois Urbana-Champaign
nhovakim@illinois.edu · (217) 244-1672

Prof. Srinivasa M. Salapaka
Professor
Mechanical Science and Engineering
University of Illinois Urbana-Champaign
salapaka@illinois.edu · (217) 244-4172

Prof. Shaunak D. Bopardikar (Ph.D. advisor)
Associate Professor
Electrical and Computer Engineering
Michigan State University
shaunak@egr.msu.edu · (517) 432-4309

Prof. Xiaofeng Wang
Associate Professor, Electrical Engineering
Molinaroli College of Engineering and Computing
University of South Carolina
wangxi@cec.sc.edu · (803) 777-9569